

Course: B. Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVOE605H Applications of Remote Sensing & GIS

Max Marks: 60

Date: 25/06/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Draw a suitable diagram wherever necessary.

Marks

Q. 1 Solve Any Two of the following.

- A) Write short notes on: - 6
- a. Remote sensing platforms
 - b. Non-selective scattering
 - c. Meta data
- B) Explain electromagnetic spectrum in detail with each spectral band's applications in remote sensing. 6
- C) Describe various important elements of visual interpretation of image data. 6

Q.2 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. Radiometric resolution
 - b. Synthetic Aperture Radar
 - c. Geostationary orbit
- B) Explain a multispectral imaging sensor system with its advantages and limitations. 6
- C) Write a detailed note on active remote sensing with its advantages and limitations. 6

Q. 3 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. Image histogram
 - b. Spectral signature of soil
 - c. Supervised classification of imageries
- B) Explain various digital image enhancement techniques. 6
- C) Write a detailed note on various methods of satellite imagery corrections. 6

Q.4 Solve Any Two of the following.

- A) Write a short note on: - **6**
- a. Geographical Information System
 - b. Triangular Irregular Network
 - c. Geocoding
- B) Explain the difference between raster data and vector data. **6**
- C) Write detailed notes on applications of DEM data with collection methods. **6**

Q. 5 Solve Any Two of the following.

- A) Write a short note on: - **6**
- a. Graticular Network
 - b. Cartesian Coordinate System
 - c. Geoid
- B) Explain in detail about cylindrical systems with types and their limitations. **6**
- C) Write a detailed note on “Projected Coordinate System” with its pros and cons. **6**

***** End *****